In [1]:
```
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
pd.read_csv(r"C:\Users\USER\Documents\Datasets downloads here\churn-bigml-20.csv")
```

Out[2]:

	State	Account length	Area code	International plan	Voice mail plan	Number vmail messages	Total day minutes	Total day calls	Total day charge	Tot ev minut
0	LA	117	408	No	No	0	184.5	97	31.37	351
1	IN	65	415	No	No	0	129.1	137	21.95	228
2	NY	161	415	No	No	0	332.9	67	56.59	317
3	SC	111	415	No	No	0	110.4	103	18.77	137
4	HI	49	510	No	No	0	119.3	117	20.28	215
...	...	...	...	...	...	...	...	...	...	
662	WI	114	415	No	Yes	26	137.1	88	23.31	155
663	AL	106	408	No	Yes	29	83.6	131	14.21	203
664	VT	60	415	No	No	0	193.9	118	32.96	85
665	WV	159	415	No	No	0	169.8	114	28.87	197
666	CT	184	510	Yes	No	0	213.8	105	36.35	159

667 rows × 20 columns

In [3]:
```
test=pd.read_csv(r"C:\Users\USER\Documents\Datasets downloads here\churn-bigml-20.c
```

In [4]:
```
pd.read_csv(r"C:\Users\USER\Documents\Datasets downloads here\churn-bigml-80.csv")
```

Out[4]:

	State	Account length	Area code	International plan	Voice mail plan	Number vmail messages	Total day minutes	Total day calls	Total day charge	To(minu
0	KS	128	415	No	Yes	25	265.1	110	45.07	19
1	OH	107	415	No	Yes	26	161.6	123	27.47	19
2	NJ	137	415	No	No	0	243.4	114	41.38	12
3	OH	84	408	Yes	No	0	299.4	71	50.90	6
4	OK	75	415	Yes	No	0	166.7	113	28.34	14
...	...	...	...	...	...	...	...	...	...	
2661	SC	79	415	No	No	0	134.7	98	22.90	18
2662	AZ	192	415	No	Yes	36	156.2	77	26.55	21
2663	WV	68	415	No	No	0	231.1	57	39.29	15
2664	RI	28	510	No	No	0	180.8	109	30.74	28
2665	TN	74	415	No	Yes	25	234.4	113	39.85	26

2666 rows × 20 columns

In [5]:
```
train=pd.read_csv(r"C:\Users\USER\Documents\Datasets downloads here\churn-bigml-80.
```

In [6]:
```
print(train.shape)
```

```
(2666, 20)
```

In [7]:
```
print(test.shape)
```

```
(667, 20)
```

In [8]:
```
df = pd.concat([train, test], ignore_index=True)
```

In [9]:
```
print(df.shape)
```

```
(3333, 20)
```

In [10]:
```
df.head()
```

Out[10]:

	State	Account length	Area code	International plan	Voice mail plan	Number vmail messages	Total day minutes	Total day calls	Total day charge	Total eve minutes
0	KS	128	415	No	Yes	25	265.1	110	45.07	197.4
1	OH	107	415	No	Yes	26	161.6	123	27.47	195.5
2	NJ	137	415	No	No	0	243.4	114	41.38	121.2
3	OH	84	408	Yes	No	0	299.4	71	50.90	61.9
4	OK	75	415	Yes	No	0	166.7	113	28.34	148.3

In [11]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3333 entries, 0 to 3332
Data columns (total 20 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   State                   3333 non-null   object 
 1   Account length          3333 non-null   int64  
 2   Area code               3333 non-null   int64  
 3   International plan      3333 non-null   object 
 4   Voice mail plan         3333 non-null   object 
 5   Number vmail messages   3333 non-null   int64  
 6   Total day minutes       3333 non-null   float64
 7   Total day calls         3333 non-null   int64  
 8   Total day charge        3333 non-null   float64
 9   Total eve minutes       3333 non-null   float64
 10  Total eve calls         3333 non-null   int64  
 11  Total eve charge        3333 non-null   float64
 12  Total night minutes     3333 non-null   float64
 13  Total night calls       3333 non-null   int64  
 14  Total night charge      3333 non-null   float64
 15  Total intl minutes      3333 non-null   float64
 16  Total intl calls        3333 non-null   int64  
 17  Total intl charge       3333 non-null   float64
 18  Customer service calls  3333 non-null   int64  
 19  Churn                   3333 non-null   bool   
dtypes: bool(1), float64(8), int64(8), object(3)
memory usage: 498.1+ KB
None
```

In [12]:
```
print(df.describe())
```

```
       Account length    Area code  Number vmail messages  Total day minutes  \
count     3333.000000  3333.000000            3333.000000        3333.000000   
mean       101.064806   437.182418               8.099010         179.775098   
std         39.822106    42.371290              13.688365          54.467389   
min          1.000000   408.000000               0.000000           0.000000   
25%         74.000000   408.000000               0.000000         143.700000   
50%        101.000000   415.000000               0.000000         179.400000   
75%        127.000000   510.000000              20.000000         216.400000   
max        243.000000   510.000000              51.000000         350.800000   

       Total day calls  Total day charge  Total eve minutes  Total eve calls  \
count      3333.000000       3333.000000        3333.000000      3333.000000   
mean        100.435644         30.562307         200.980348       100.114311   
std          20.069084          9.259435          50.713844        19.922625   
min           0.000000          0.000000           0.000000         0.000000   
25%          87.000000         24.430000         166.600000        87.000000   
50%         101.000000         30.500000         201.400000       100.000000   
75%         114.000000         36.790000         235.300000       114.000000   
max         165.000000         59.640000         363.700000       170.000000   

       Total eve charge  Total night minutes  Total night calls  \
count       3333.000000          3333.000000        3333.000000   
mean          17.083540           200.872037         100.107711   
std            4.310668            50.573847          19.568609   
min            0.000000            23.200000          33.000000   
25%           14.160000           167.000000          87.000000   
50%           17.120000           201.200000         100.000000   
75%           20.000000           235.300000         113.000000   
max           30.910000           395.000000         175.000000   

       Total night charge  Total intl minutes  Total intl calls  \
count         3333.000000         3333.000000       3333.000000   
mean             9.039325           10.237294          4.479448   
std              2.275873            2.791840          2.461214   
min              1.040000            0.000000          0.000000   
25%              7.520000            8.500000          3.000000   
50%              9.050000           10.300000          4.000000   
75%             10.590000           12.100000          6.000000   
max             17.770000           20.000000         20.000000   

       Total intl charge  Customer service calls  
count        3333.000000             3333.000000  
mean            2.764581                1.562856  
std             0.753773                1.315491  
min             0.000000                0.000000  
25%             2.300000                1.000000  
50%             2.780000                1.000000  
75%             3.270000                2.000000  
max             5.400000                9.000000  
```

In [13]:
```
print(df.isnull().sum())
```

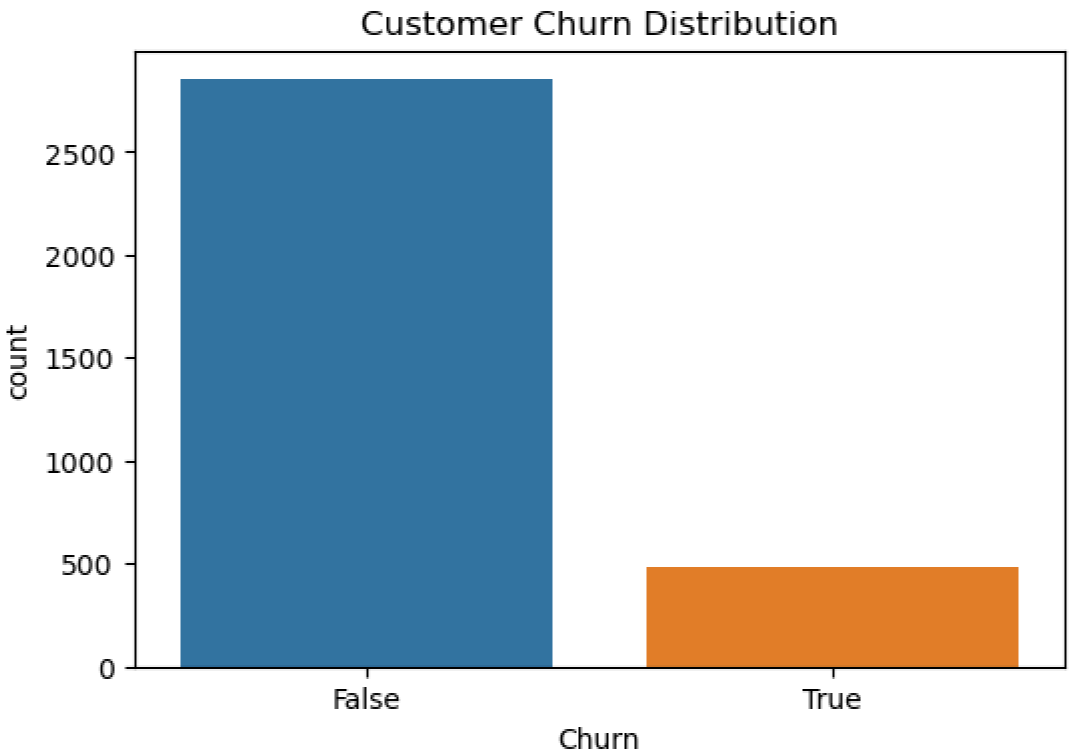
```
State                     0
Account length            0
Area code                 0
International plan        0
Voice mail plan           0
Number vmail messages     0
Total day minutes         0
Total day calls           0
Total day charge          0
Total eve minutes         0
Total eve calls           0
Total eve charge          0
Total night minutes       0
Total night calls         0
Total night charge        0
Total intl minutes        0
Total intl calls          0
Total intl charge         0
Customer service calls    0
Churn                     0
dtype: int64
```

In [14]:
```
print(df['Churn'].value_counts())
```

```
Churn
False    2850
True      483
Name: count, dtype: int64
```

In [15]:
```
churn_counts = df['Churn'].value_counts()

plt.figure(figsize=(6,4))
sns.countplot(x='Churn', data=df)
plt.title("Customer Churn Distribution")
plt.show()
```



Business Insight:

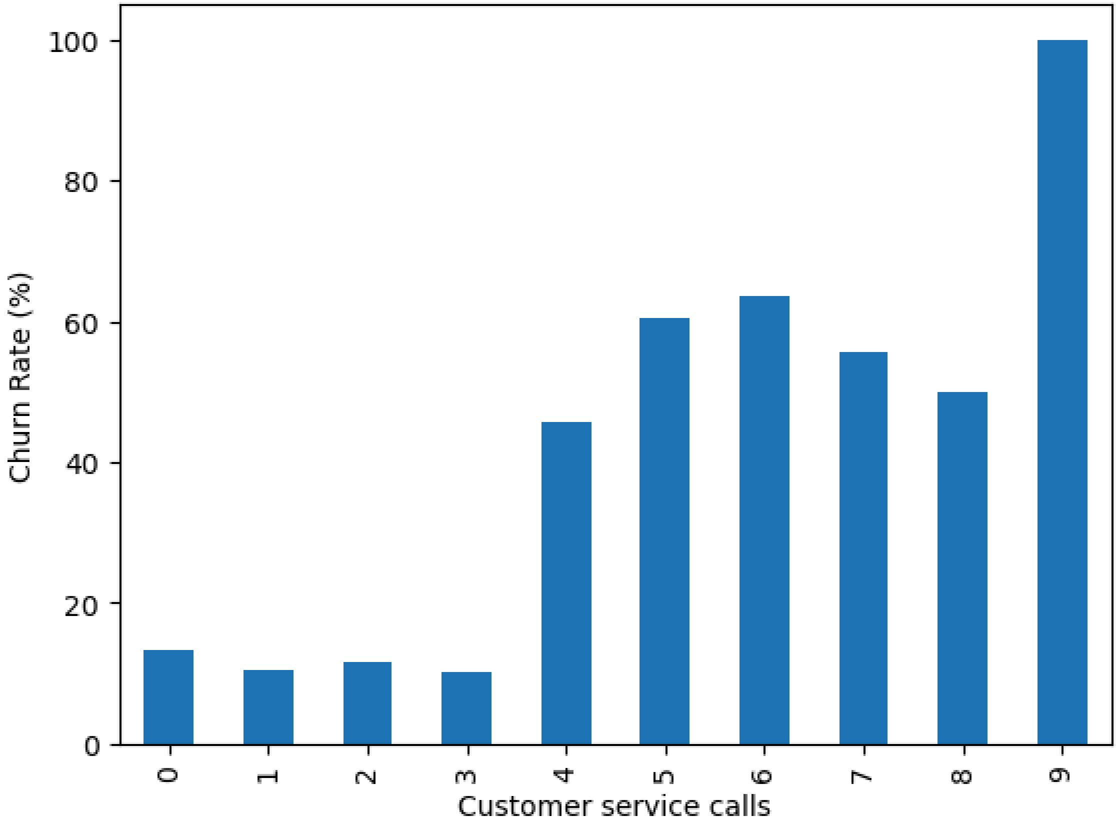
From the chart:

Approximately 2,850 customers remained with the company. Approximately 480 customers churned (left the company).

Recommendation: Develop a churn prediction model and implement targeted retention programs for high-risk customers to reduce customer attrition and improve long-term profitability.

In [16]:
```
service_calls = df.groupby(
    'Customer service calls'
)['Churn'].mean()*100

service_calls.plot(kind='bar')
plt.ylabel("Churn Rate (%)")
plt.show()
```



insights: Customers who made 0–3 service calls have a relatively low churn rate (around 10–14%).

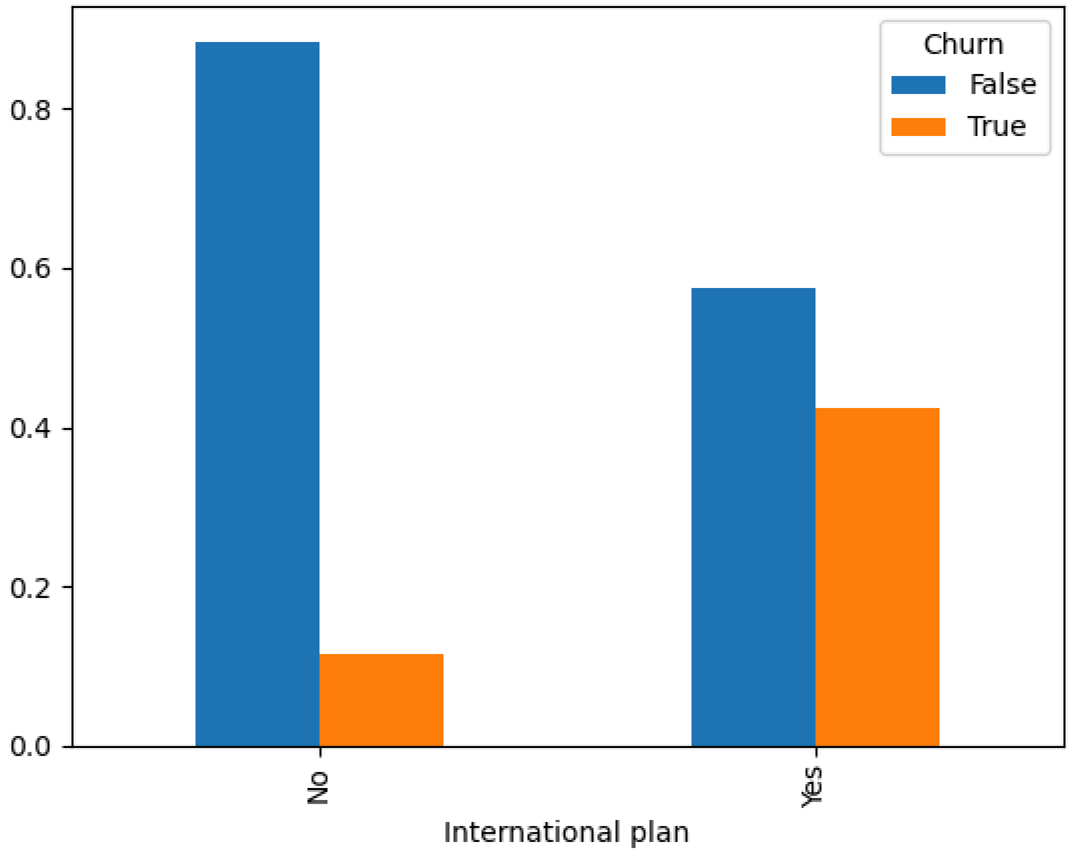
Customers with 4 or more service calls are significantly more likely to leave than those with fewer calls.

This suggests that repeated contact with customer support may be a sign of unresolved issues, poor service experience, billing disputes, network problems, or dissatisfaction.

recommendation:Implement a proactive retention strategy that automatically flags customers with four or more service calls and prioritizes them for resolution and engagement efforts. This can reduce churn and improve customer satisfaction.

In [17]:
```
pd.crosstab(
    df['International plan'],
    df['Churn'],
    normalize='index'
).plot(kind='bar')
```

Out[17]: <Axes: xlabel='International plan'>



Insights:This chart compares churn behavior between customers who have an international plan and those who do not.

Key Findings

- Customers without an international plan:

About 89% stayed with the company. About 11% churned.

- Customers with an international plan:

About 58% stayed. About 42% churned.

-Customers with an international plan are almost 4 times more likely to churn than customers without one.

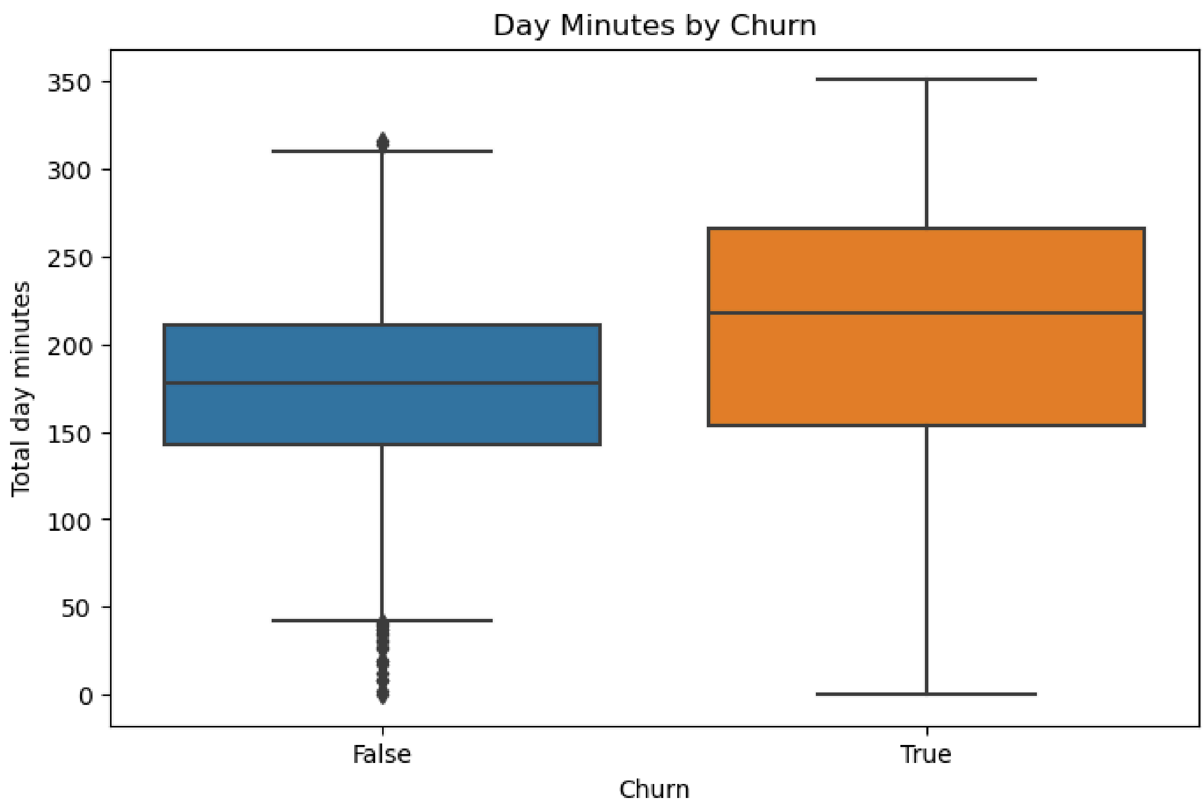
recommendations:Review international plan pricing and service performance, while implementing targeted retention strategies for international-plan customers. Combining international-plan status with customer service call data can help identify customers at the highest risk of churn.

In [18]:
```
#Day Minutes Usage
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Churn',
```

```
    y='Total day minutes',
    data=df
)

plt.title("Day Minutes by Churn")
plt.show()
```



insights: This boxplot compares the distribution of Total Day Minutes for customers who stayed (False) versus those who churned (True).

Key Findings

- Customers who stayed:

Median day usage ≈ 180 minutes Most customers fall between 145–210 minutes

-Customers who churned:

Median day usage ≈ 220 minutes Most customers fall between 155–265 minutes

-Churned customers consistently use more daytime minutes than retained customers.

The median usage of churned customers is about 40 minutes higher than non-churned customers.

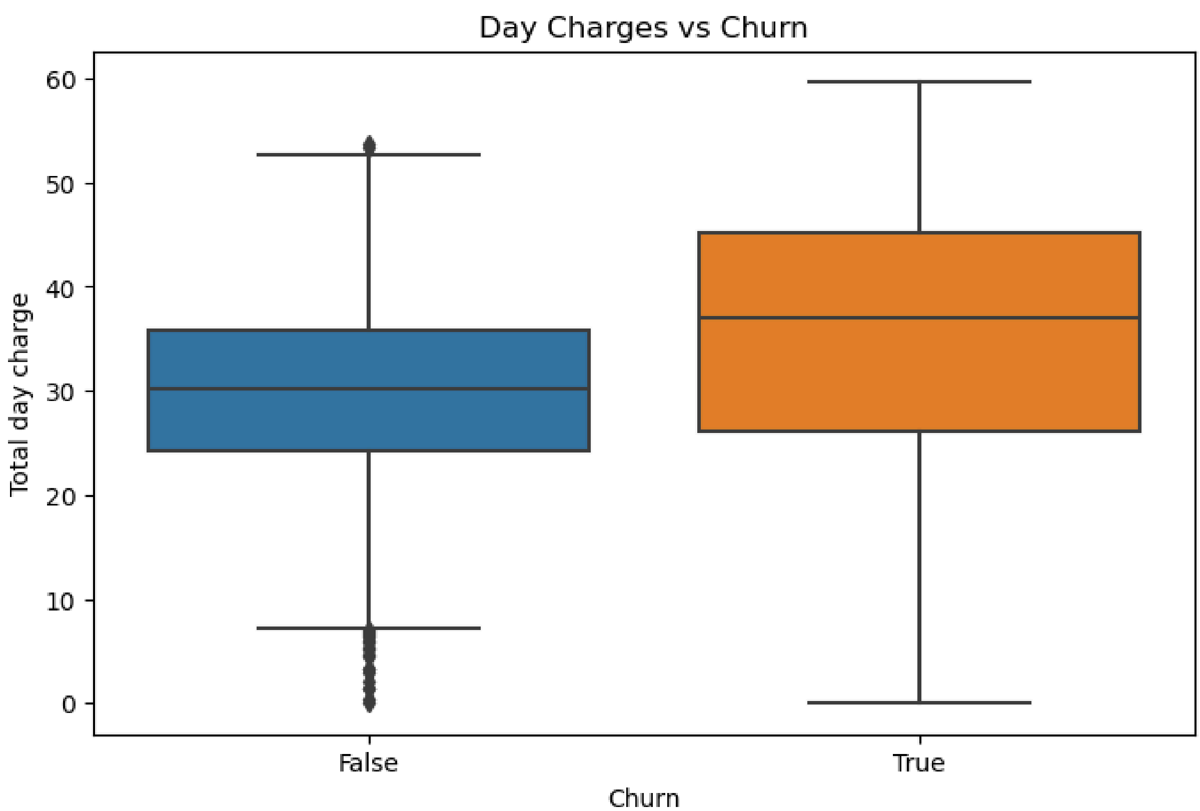
recommendations:Develop retention strategies specifically for high-usage customers, including customized pricing plans, loyalty incentives, and proactive customer engagement programs to reduce churn among this valuable segment. This graph is showing something management cares about:

The customers generating the most usage (and likely the most revenue) are the ones most likely to churn.

In [24]:
```
#Day Charges Impact
plt.figure(figsize=(8,5))

sns.boxplot(
    x='Churn',
    y='Total day charge',
    data=df
)

plt.title("Day Charges vs Churn")
plt.show()
```



insights:The box plot compares Total Day Charge for customers who did not churn (False) and those who churned (True).

Insight 1: Customers with higher day charges are more likely to churn Customers who churned have a higher median day charge (around $37)comparedtocustomerswhostayed(around$30). This suggests that customers paying more for daytime usage are more likely to leave the company.

Business Meaning: High daytime charges may be causing customer dissatisfaction, making customers seek cheaper alternatives from competitors.

Insight 2: Churn customers show greater variability in day charges The spread (IQR) for churned customers is much wider. Some churned customers have extremely high day charges, reaching nearly $60.

Business Meaning: Heavy daytime users appear to have different spending patterns and may be more sensitive to pricing.

Insight 3: Non-churn customers have more consistent day charges Customers who remained have a tighter distribution centered around $25–35. Their spending behavior is more stable.

Business Meaning: Moderate users appear to be more satisfied with the service and pricing.

Insight 4: Several low-charge outliers exist among non-churn customers There are a few customers with very low day charges who still remained loyal.

Business Meaning: Low usage alone does not increase churn risk. Other factors such as customer service, international plans, or support interactions likely influence retention.

Business Recommendations

1. Introduce pricing plans for heavy daytime users

Since high day charges are associated with higher churn:

Offer discounted daytime bundles. Introduce unlimited daytime calling plans. Reward loyal high-usage customers with special pricing.

Expected Impact:

Lower churn among high-value customers. Increase customer lifetime value. 2. Identify high-risk customers early

Build a monitoring system that flags customers with:

High total day minutes High day charges Increasing monthly bills

These customers can receive:

Personalized discounts Retention offers Loyalty rewards 3. Create customer segments based on usage

Segment customers into:

Low users Moderate users Heavy daytime users

Each segment should receive pricing and promotions tailored to their usage patterns.

4. Investigate additional churn drivers

Day charges alone are unlikely to fully explain churn. Analyze how they interact with:

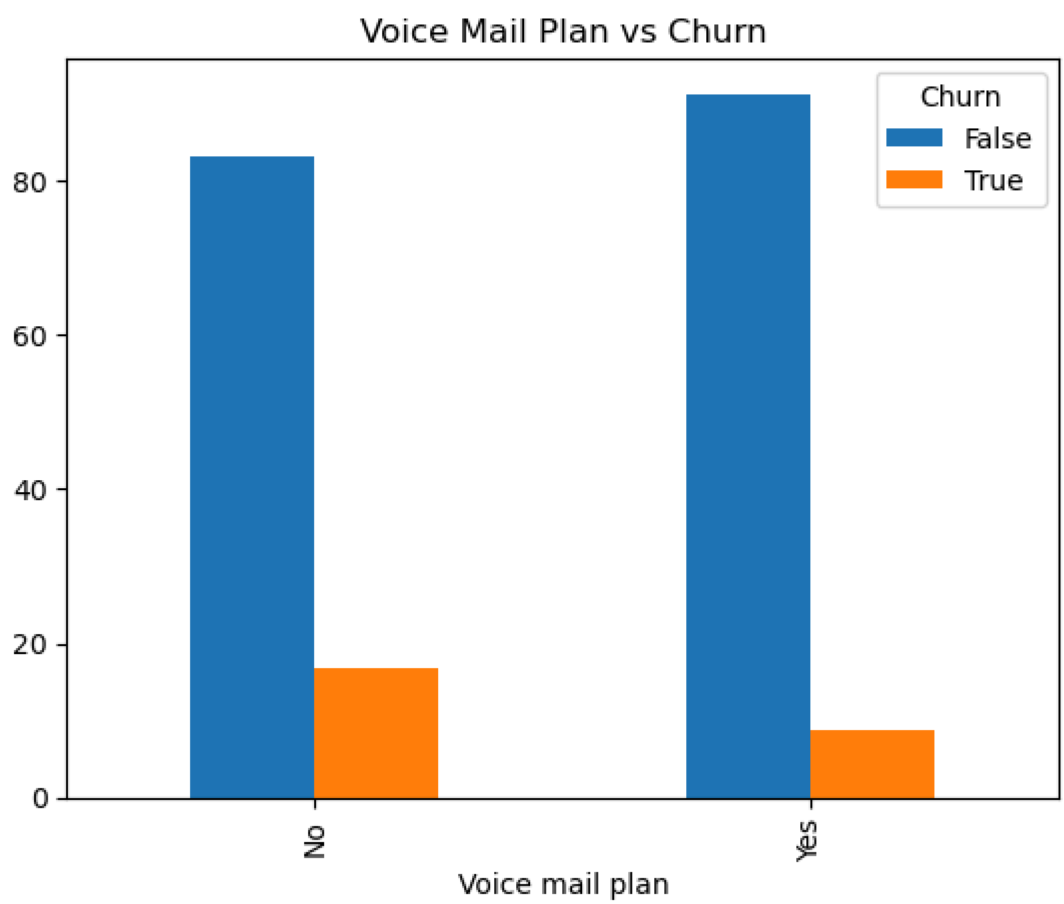
Customer service calls International plan subscription Voice mail plan Total day minutes Monthly charges

A multivariate analysis or predictive model can identify the strongest combined predictors of churn.

In [20]:
```
#Voice Mail Plan
vm = pd.crosstab(
    df['Voice mail plan'],
    df['Churn'],
    normalize='index'
)*100

vm.plot(kind='bar')

plt.title("Voice Mail Plan vs Churn")
plt.show()
```



insights: This chart compares the percentage of customers with and without a Voice Mail Plan across churned and non-churned customers.

Insight 1: Customers with a Voice Mail Plan are less likely to churn Among customers who did not churn, a larger proportion have a Voice Mail Plan (Yes) than those without one. Among customers who churned, the majority do not have a Voice Mail Plan.

Business Meaning: Having a Voice Mail Plan appears to be associated with higher customer retention. Customers using additional services may perceive greater value from their subscription.

Insight 2: Customers without a Voice Mail Plan have a higher churn rate The churn percentage is noticeably higher for customers who do not subscribe to the Voice Mail Plan. This suggests these customers may be less engaged with the company's services.

Business Meaning: Customers who only use basic services may be more willing to switch to competitors because they have fewer value-added features keeping them loyal.

Insight 3: Value-added services improve customer loyalty

The Voice Mail Plan can be viewed as a value-added feature that increases customer engagement and satisfaction.

Business Meaning: Customers using additional services tend to have stronger relationships with the company, reducing their likelihood of leaving.

Insight 4: Voice Mail Plan can serve as an indicator of churn risk

Customers without a Voice Mail Plan may represent a higher-risk segment and should be monitored more closely.

Business Meaning: The presence or absence of a Voice Mail Plan can be incorporated into predictive churn models to identify customers who may need retention efforts.

Business Recommendations

1. Promote Voice Mail Plan adoption

Offer free trial periods for the Voice Mail Plan. Bundle it with popular service packages. Highlight its benefits during customer onboarding and renewals.

Expected Impact: Higher adoption may improve customer engagement and reduce churn.

2. Target customers without a Voice Mail Plan

Develop retention campaigns for customers who do not subscribe by offering:

Discounted upgrades Personalized service bundles Loyalty incentives

Expected Impact: Encouraging these customers to adopt additional services may strengthen retention.

3. Bundle value-added services

Instead of marketing the Voice Mail Plan alone, combine it with:

Unlimited calling packages Data plans International calling options Customer loyalty rewards

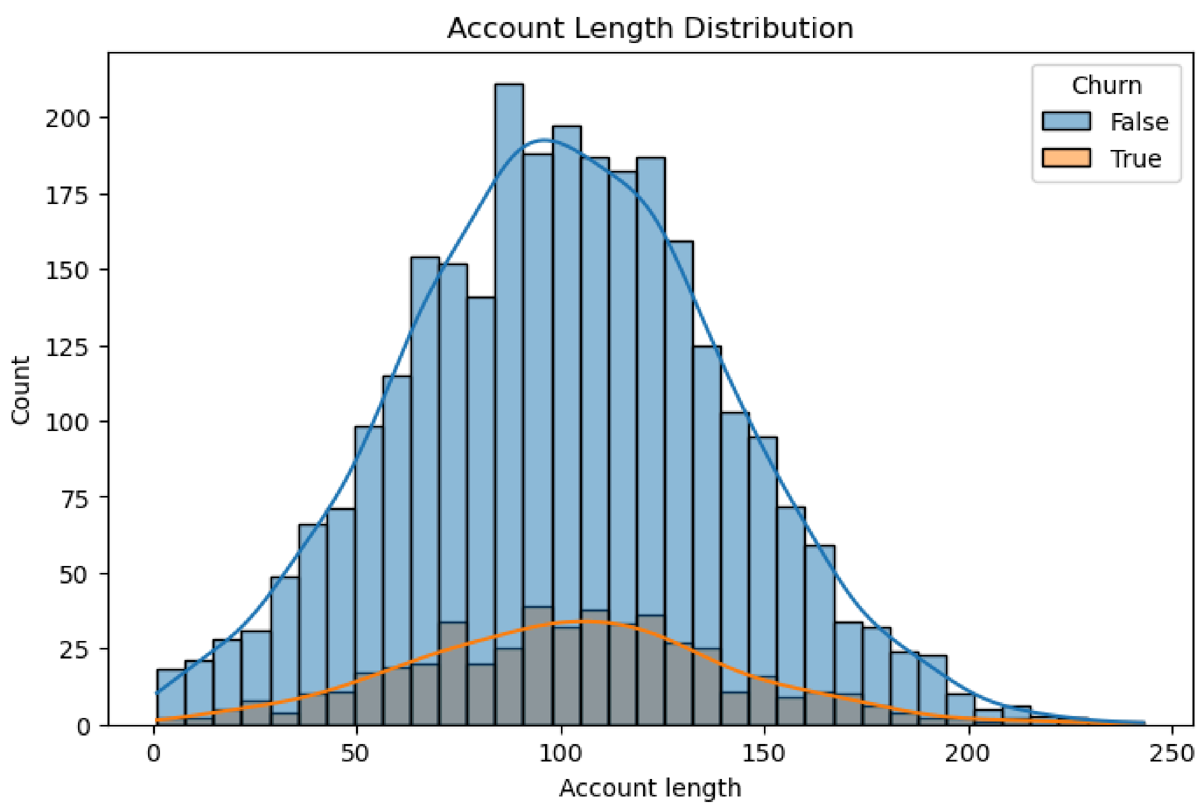
This creates a more attractive overall offering and increases switching costs.

In [22]:
```
#Account Length Distribution
plt.figure(figsize=(8,5))

sns.histplot(
    data=df,
    x='Account length',
    hue='Churn',
    kde=True
)

plt.title("Account Length Distribution")
plt.show()
```

```
C:\Users\USER\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119: FutureWarning: u
se_inf_as_na option is deprecated and will be removed in a future version. Convert i
nf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
```



insights:This histogram compares the distribution of Account Length for customers who did not churn (False) and those who churned (True).

Insight 1: Churn occurs across all account lengths Customers who churn are distributed across nearly the entire range of account lengths. There is no single account age where churn is concentrated.

Business Meaning: Both new and long-term customers are at risk of leaving, indicating that account tenure alone is not a strong predictor of churn.

Insight 2: Most customers have account lengths between 80 and 120 days Both churned and retained customers are heavily concentrated around 80–120 days. This suggests the customer base is centered around medium-tenure accounts.

Business Meaning: Since both groups share a similar distribution, account length by itself does not clearly differentiate loyal customers from those who churn.

Insight 3: Similar distribution between churned and retained customers The shapes of the churn and non-churn distributions largely overlap. There is no obvious shift toward shorter or longer account lengths among customers who churn.

Business Meaning: Unlike features such as customer service calls or total day charges, account length has limited standalone explanatory power for churn.

Insight 4: Account length should be combined with other variables While account length alone is not a strong indicator, it may become more informative when analyzed alongside variables such as: Customer service calls Monthly/day charges International plan Voice mail plan Total day minutes

Business Meaning: Interactions between tenure and customer behavior may reveal hidden churn patterns that are not visible from account length alone.

Business Recommendations

1. Do not rely on account length alone for churn prediction

Use account length as one feature within a broader predictive model rather than as the primary driver of retention decisions.

Expected Impact: Improves prediction accuracy by focusing on multiple customer behaviors rather than tenure alone.

2. Personalize engagement throughout the customer lifecycle

Since churn occurs among both newer and longer-tenured customers:

Welcome and educate new customers. Regularly engage mid-tenure customers with relevant offers. Reward long-term customers through loyalty programs and exclusive benefits.

Expected Impact: Maintains customer satisfaction across all stages of the customer journey.

3. Combine tenure with behavioral indicators

Develop customer segments based on:

Account length Usage patterns Billing amounts Customer service interactions

This allows the company to identify customers who are both long-tenured and showing signs of dissatisfaction.

4. Focus retention efforts on behavioral risk factors

Prioritize proactive interventions for customers exhibiting:

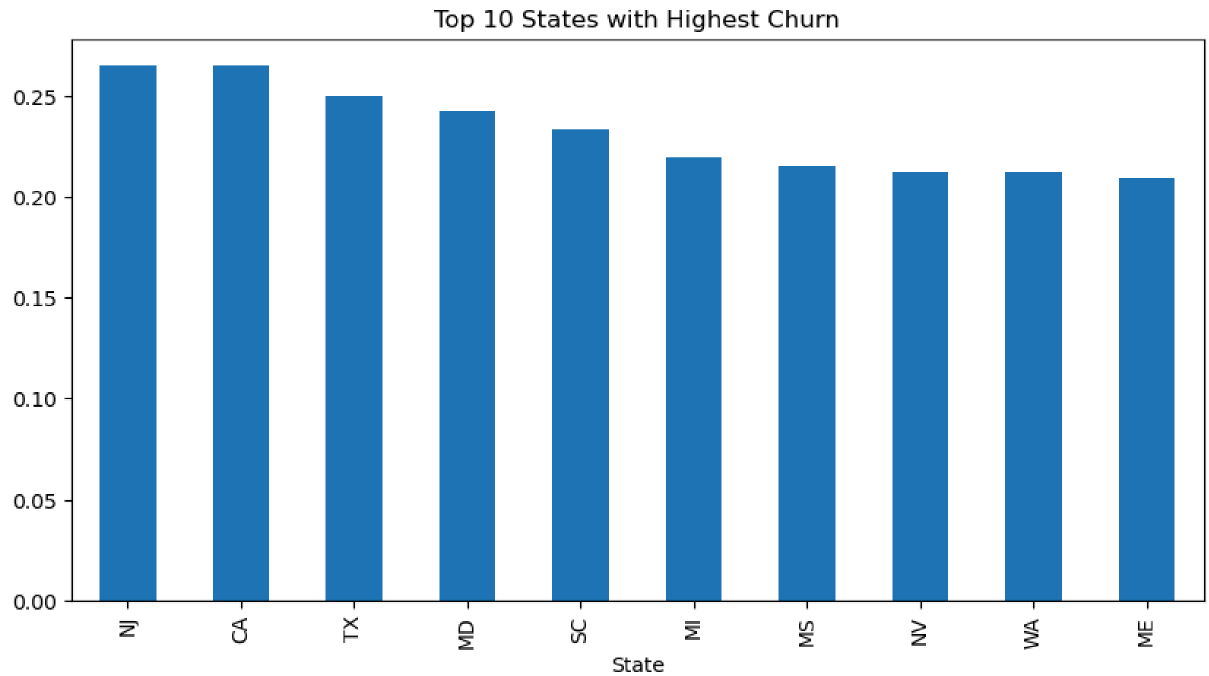
High day charges Frequent customer service calls International plan usage No voice mail plan

These factors are likely to be stronger predictors of churn than account length alone.

In [22]:
```
#top states hughest churn
state_churn = (
    df.groupby('State')['Churn']
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

state_churn.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title("Top 10 States with Highest Churn")
plt.show()
```



insights:This bar chart highlights the top 10 states with the highest customer churn rates. The highest churn rates are observed in New Jersey (NJ) and California (CA), followed by Texas (TX), Maryland (MD), South Carolina (SC), Michigan (MI), Mississippi (MS), Nevada (NV), Washington (WA), and Maine (ME).

Insight 1: New Jersey and California experience the highest churn rates New Jersey (NJ) and California (CA) have the highest churn rates, approximately 26–27%. This means nearly 1 in 4 customers in these states discontinue their service.

Business Meaning: These states should be prioritized for customer retention efforts, as reducing churn here could have a significant impact on overall revenue.

Insight 2: Churn is concentrated in specific geographic regions The top 10 states all have churn rates above 20%, indicating that customer attrition is not evenly distributed across the country. Geographic factors such as market competition, pricing, network quality, or customer expectations may influence churn.

Business Meaning: Regional differences suggest that a one-size-fits-all retention strategy may not be effective.

Insight 3: The difference among the top states is relatively small Churn rates range from approximately 21% to 27%. Although NJ and CA lead, the remaining states also have consistently high churn.

Business Meaning: Several states represent high-risk markets and should be included in retention planning rather than focusing only on the top one or two.

Insight 4: State-level churn may reflect local market conditions

High churn in these states may be associated with:

Stronger competition from other telecom providers. Customer dissatisfaction with pricing or service quality. Differences in customer demographics and usage behavior.

Business Meaning: Understanding local market conditions can help explain why customers in certain states are more likely to leave.

Business Recommendations

1. Prioritize retention campaigns in high-churn states

Launch targeted retention initiatives in:

New Jersey California Texas Maryland

These campaigns could include:

Loyalty rewards Personalized discounts Contract renewal incentives Exclusive service bundles

Expected Impact: Reducing churn in these states can significantly improve customer retention and revenue.

2. Investigate state-specific churn drivers

Conduct deeper analyses to determine whether churn is driven by:

High monthly or day charges Frequent customer service interactions Network coverage issues Competitor promotions Customer demographics

Expected Impact: Identifying the root causes enables more effective, localized solutions.

3. Implement regional pricing and marketing strategies

Rather than applying the same offers nationwide:

Tailor promotions based on local customer needs. Adjust pricing where competition is strongest. Highlight services that address regional preferences.

Expected Impact: More relevant offers can improve customer satisfaction and reduce churn.

4. Monitor churn trends by state

Develop an interactive dashboard that tracks:

Monthly churn rate by state Customer acquisition and retention Revenue impact Effectiveness of retention campaigns

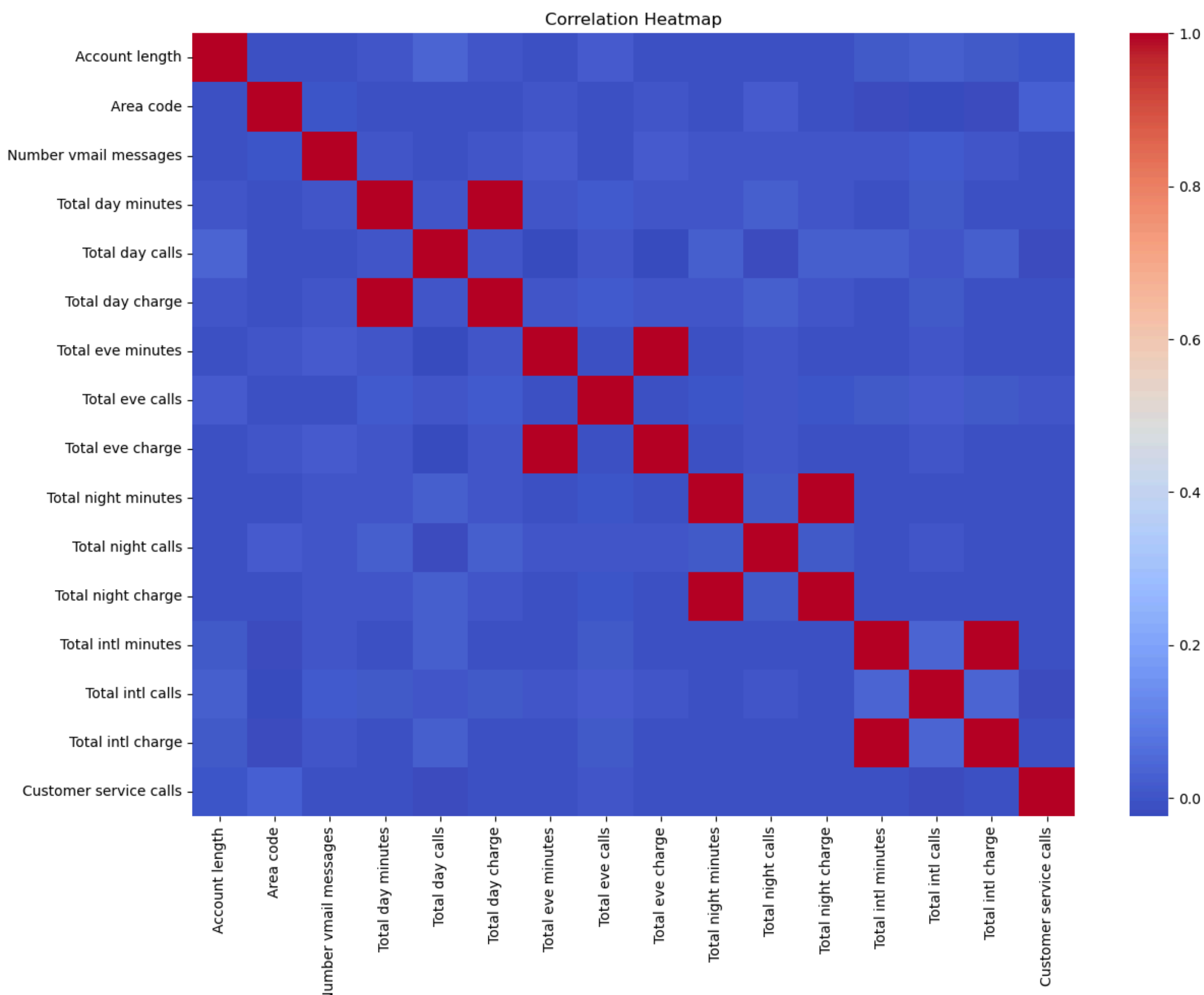
Expected Impact: Provides early warning of emerging churn hotspots and supports data-driven decision-making.

In [24]:
```
#correlation analysis
numeric_df = df.select_dtypes(
    include=['int64','float64']
)

plt.figure(figsize=(14,10))

sns.heatmap(
    numeric_df.corr(),
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")
plt.show()
```



insigts:The correlation heatmap shows the relationships between the numerical variables in the telecom customer dataset. Correlation values range from -1 to +1, where values closer to +1 indicate a strong positive relationship, values near 0 indicate little or no relationship, and values closer to -1 indicate a strong negative relationship.

Insight 1: Minutes and Charges are perfectly correlated

The strongest relationships are observed between:

Total Day Minutes ↔ Total Day Charge Total Evening Minutes ↔ Total Evening Charge Total Night Minutes ↔ Total Night Charge Total International Minutes ↔ Total International Charge

These pairs show an almost perfect positive correlation (≈1.0).

Business Meaning: Charges are directly calculated from call duration. As customers use more minutes, their charges increase proportionally.

Recommendation:

For predictive modeling, keep either the minutes or the charges, but not both. Removing one variable from each pair reduces multicollinearity, improving model stability and interpretability. Insight 2: Number of calls has little relationship with charges

Variables such as:

Total Day Calls Total Evening Calls Total Night Calls Total International Calls

have weak correlations with their corresponding charges.

Business Meaning: Customers' bills depend more on how long they talk rather than how many calls they make. A customer making fewer but longer calls may incur higher charges than one making many short calls.

Recommendation: Focus pricing strategies and customer segmentation on call duration rather than call frequency.

Insight 3: Customer Service Calls are largely independent

The Customer Service Calls feature has very weak correlations with most numerical variables.

Business Meaning: Customers contact support for reasons beyond usage, such as:

Billing issues Network problems Technical support Service dissatisfaction

This suggests customer service interactions capture a different dimension of customer behavior.

Recommendation: Although it has low correlation with usage metrics, Customer Service Calls should be retained in churn prediction models because it often has strong predictive value for identifying dissatisfied customers.

Insight 4: Account Length has minimal correlation with usage metrics

Account Length shows weak correlations with minutes, charges, and calls.

Business Meaning: The length of time a customer has been with the company does not strongly influence how much they use telecom services.

Recommendation: Use Account Length alongside behavioral variables rather than relying on it as a standalone predictor.

Insight 5: Most variables are weakly correlated

Apart from the expected minutes–charges relationships, the remaining variables exhibit weak correlations.

Business Meaning: Each feature provides distinct information about customer behavior, reducing concerns about redundancy.

Recommendation: This is beneficial for machine learning because it allows the model to learn from multiple independent aspects of customer behavior.

Business Recommendations

1. Remove redundant variables before modeling

Since charges are derived directly from minutes:

Keep only minutes or charges from each category (day, evening, night, international). This reduces redundancy and improves model efficiency. 2. Prioritize behavioral metrics

Features such as:

Total day minutes Total international minutes Customer service calls Voice mail plan International plan

are likely to contribute more meaningful information to churn prediction than simple account demographics.

3. Use feature selection techniques

Apply methods such as:

Correlation analysis Recursive Feature Elimination (RFE) Feature importance from Random Forest or XGBoost L1 (Lasso) regularization

These techniques help identify the most influential variables while reducing unnecessary complexity.

In [23]:
```
#high risk customers
high_risk = df[
    (df['Customer service calls'] >= 4)
    &
    (df['International plan'] == 'Yes')
]

print(high_risk.shape)
```

```
high_risk.head()
```

```
(28, 20)
```

Out[23]:

	State	Account length	Area code	International plan	Voice mail plan	Number vmail messages	Total day minutes	Total day calls	Total day charge	Tot ev minut
146	WV	94	510	Yes	Yes	23	197.1	125	33.51	214
162	ME	131	510	Yes	Yes	26	292.9	101	49.79	199
286	AR	179	415	Yes	Yes	38	220.1	78	37.42	234
435	MN	152	415	Yes	Yes	20	237.5	120	40.38	253
940	MD	88	415	Yes	No	0	235.1	98	39.97	251

insights CustomerChurn Risk Analysis Business Recommendation 146 (WV) No Has an International Plan, Voice Mail Plan, high day usage (197.1 mins), and 4 customer service calls but remained loyal. Monitor closely. Although this customer has not churned, the high number of service calls indicates potential dissatisfaction. Offer proactive support and loyalty incentives. 162 (ME) Yes Very high day usage (292.9 mins), highest day charge ($49.79), International Plan, 4 customer service calls. This customer likely left due to high usage costs or dissatisfaction. Offer discounted unlimited daytime plans or personalized pricing to similar customers before they churn. 286 (AR) No High usage across all periods but moderate international usage and only 4 customer service calls. This is a valuable customer. Introduce loyalty rewards and premium packages to maintain retention. 435 (MN) Yes High day, evening, and international usage with 9 customer service calls, the highest in the sample. This customer showed strong signs of dissatisfaction. Customers with frequent support interactions should be escalated immediately to retention teams before they decide to leave. 940 (MD) Yes High day and evening charges, no Voice Mail Plan, 4 customer service calls. Customers without value-added services may perceive less value. Promote bundled packages and investigate the reasons behind repeated service calls.

1. Customer Service Calls are a strong churn indicator

Three of the five customers contacted customer service four or more times. One churned customer contacted support nine times, indicating unresolved issues.

Recommendation

Automatically flag customers after three or more service calls. Assign them to a specialized retention or customer success team. 2. High day usage is associated with churn

The churned customers have some of the highest day minutes and day charges in the sample.

Recommendation

Offer unlimited daytime calling packages. Provide personalized discounts for heavy users before they become dissatisfied. 3. Voice Mail Plan may improve retention

Among the churned customers:

One customer did not have a Voice Mail Plan. Customers with additional services generally appear more engaged.

Recommendation

Bundle value-added services such as Voice Mail with premium plans to increase perceived value and customer loyalty. 4. International Plan customers require closer monitoring

Every customer in this sample has an International Plan, yet several still churned.

Recommendation

Evaluate whether international pricing is competitive. Introduce discounted international calling packages or flexible international bundles. 5. High-value customers are also at risk

The churned customers are not low-usage customers—they generate substantial revenue through high usage.

Recommendation

Focus retention efforts on high-value customers, as retaining them has a greater financial impact than acquiring new customers.

In [25]:
```
from sklearn.ensemble import RandomForestClassifier

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import (classification_report,confusion_matrix,roc_auc_score,
```

In [26]:
```
le = LabelEncoder()

for col in ["State", "International plan", "Voice mail plan"]:

    df[col] = le.fit_transform(df[col])
```

In [27]:
```
X = df.drop("Churn", axis=1)

y = df["Churn"].astype(int)
```

In [30]:
```
train_size = len(train)
X_train, X_test = X.iloc[:train_size], X.iloc[train_size:]

y_train, y_test = y.iloc[:train_size], y.iloc[train_size:]
print(f"\nTrain : {X_train.shape}   Test : {X_test.shape}")
```

```
Train : (2666, 19)   Test : (667, 19)
```

In [32]:
```python
model = RandomForestClassifier(n_estimators=100, random_state=42,class_weight="bala

model.fit(X_train, y_train)
```

Out[32]:
```
▾ RandomForestClassifier

RandomForestClassifier(class_weight='balanced', random_state=42)
```

In [34]:
```python
y_pred  = model.predict(X_test)

y_proba = model.predict_proba(X_test)[:, 1]
```

In [35]:
```python
print("\n=== Classification Report ===")

print(classification_report(y_test, y_pred, target_names=["No Churn", "Churn"]))



print(f"Accuracy : {accuracy_score(y_test, y_pred):.4f}")

print(f"ROC-AUC  : {roc_auc_score(y_test, y_proba):.4f}")



print("\n=== Confusion Matrix ===")

print(confusion_matrix(y_test, y_pred))
```

```
=== Classification Report ===
              precision    recall  f1-score   support

    No Churn       0.95      1.00      0.97       572
       Churn       0.99      0.69      0.81        95

    accuracy                           0.96       667
   macro avg       0.97      0.85      0.89       667
weighted avg       0.96      0.96      0.95       667

Accuracy : 0.9550
ROC-AUC  : 0.9157

=== Confusion Matrix ===
[[571   1]
 [ 29  66]]
```

In [ ]: